

# Systems Under Pressure

## Shared Group Investigation Notebook

DRIVING QUESTION

How do we know when a system is actually in danger?

GROUP DETAILS

GROUP NAME \_\_\_\_\_

MEMBERS \_\_\_\_\_  
\_\_\_\_\_

INVESTIGATION TOPIC \_\_\_\_\_

CATEGORY \_\_\_\_\_

ESSENTIAL QUESTION \_\_\_\_\_

HOW THIS NOTEBOOK WORKS

- One shared investigation journal per group.
- Individual handwritten extensions are separate and shown during vlogs.
- Three informal vlogs: sequences, exponentials, polynomials.
- No daily vlogs. No required 150-word reflections.

CORE ROUTINES

01 SYSTEM -> VARIABLES -> MODEL -> CRITIQUE

02 THINK FIRST -> SPARKY -> VERIFY -> DECIDE

03 LOCAL CHECK vs GLOBAL CHECK

04 WHAT THE MODEL REVEALS / HIDES

05 ONE REJECTED MODEL

THE WORK OF THIS NOTEBOOK

Graphs, equations, calculations, annotations, revision, evidence, and one rejected model.

# Making Sense of Our Topic

Before building any model, understand the situation: who is affected, what they experience, what the data shows, and what it hides.

|         |           |          |     |
|---------|-----------|----------|-----|
| OBSERVE | EMPATHISE | QUESTION | MAP |
|---------|-----------|----------|-----|

CLASS 1 LAUNCH FOCUS

Empathy informs which variables matter and which thresholds are meaningful to real people, not just to a graph.

Use this page for free exploration: what do we know, what surprises us, what makes this urgent, and what investigation question should we draft?

WHAT HAPPENS IN CLASS

- > Experience the Bogotá water crisis launch: observe before analysing.
- > Identify who is affected and how.
- > Discuss what the crisis graph shows and cannot show.
- > Choose your investigation category and begin a group question.

OPTIONAL SPARKY SUPPORT

- > What are 5 very different ways experts think about our topic?
- > Who benefits when this system is healthy? Who suffers most when it fails?
- > What would someone who disagrees argue, and what evidence would they use?

GROUP PRODUCES

- > Initial sense of the topic and draft investigation question.
- > At least 2 stakeholder perspectives mapped.
- > Connected systems map with one confounding variable named.
- > One possible source or graph identified.

OPEN WORKSPACE: notice, question, sketch, list, connect

Use this page to identify who is affected, what they experience, what variables they would track, and where perspectives conflict.

| SYSTEM                                                       | VARIABLES                                                                     | MODEL                                                          | CRITIQUE |
|--------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------|----------|
| <div>Stakeholders</div> <div>Who is affected, and how?</div> | <div>Variables</div> <div>What would each person measure or care about?</div> | <div>Conflict</div> <div>Where do perspectives disagree?</div> |          |
| OPEN WORKSPACE                                               |                                                                               |                                                                |          |

## Variables + Hidden Systems

Use this page to map connected variables, possible hidden pressures, measurable evidence, and at least one confounding variable.

| SYSTEM                                                                 | VARIABLES                                                             | MODEL                                                                       | CRITIQUE |
|------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------|----------|
| <div>System Map</div> <div>What connected pressures are visible?</div> | <div>Measurable</div> <div>What can be measured with real data?</div> | <div>Hidden Pressure</div> <div>What might be missing or confounding?</div> |          |
| OPEN WORKSPACE                                                         |                                                                       |                                                                             |          |

# Graph Observations

Use this page to annotate trends, thresholds, anomalies, and intervals before choosing equations.

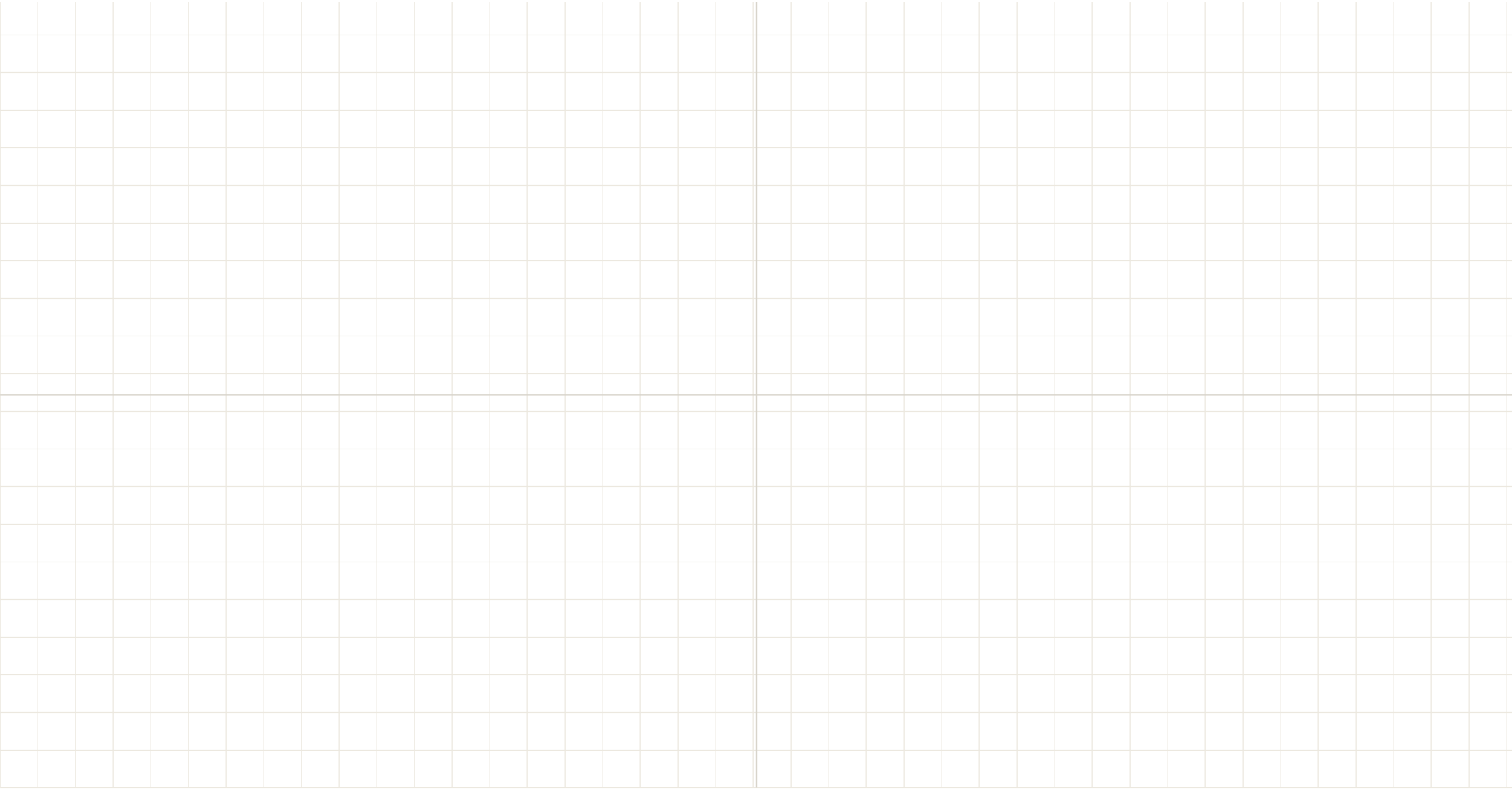
| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

Graph First  
What feature matters most?

Interval  
Where does the pattern change?

Evidence  
Which part supports your claim?

OPEN WORKSPACE



# Local vs Global Behavior

Use this page to compare a short interval with a longer interval and notice model failure.

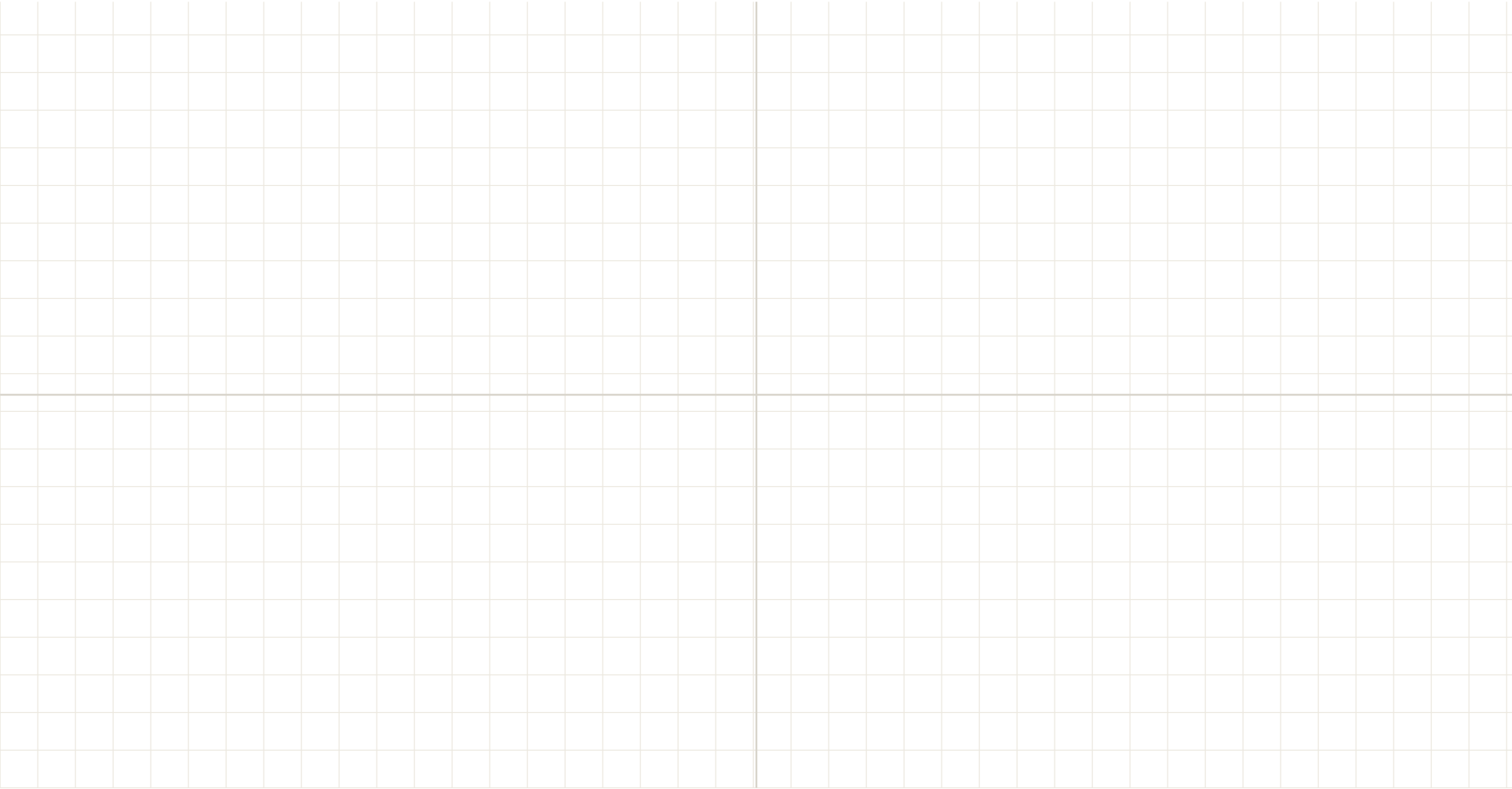
| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

Zoom In  
What seems true here?

Zoom Out  
Does it still make sense globally?

Bad Extrapolation  
Where did the model stop being meaningful?

OPEN WORKSPACE



# Sequences + Series Modeling

Use this page to model repeated change, accumulation, prediction, and where the pattern breaks.

| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

Recursive  
How does each term depend on the one before it?

recursive  $a_n = r \cdot a_{n-1} + d$

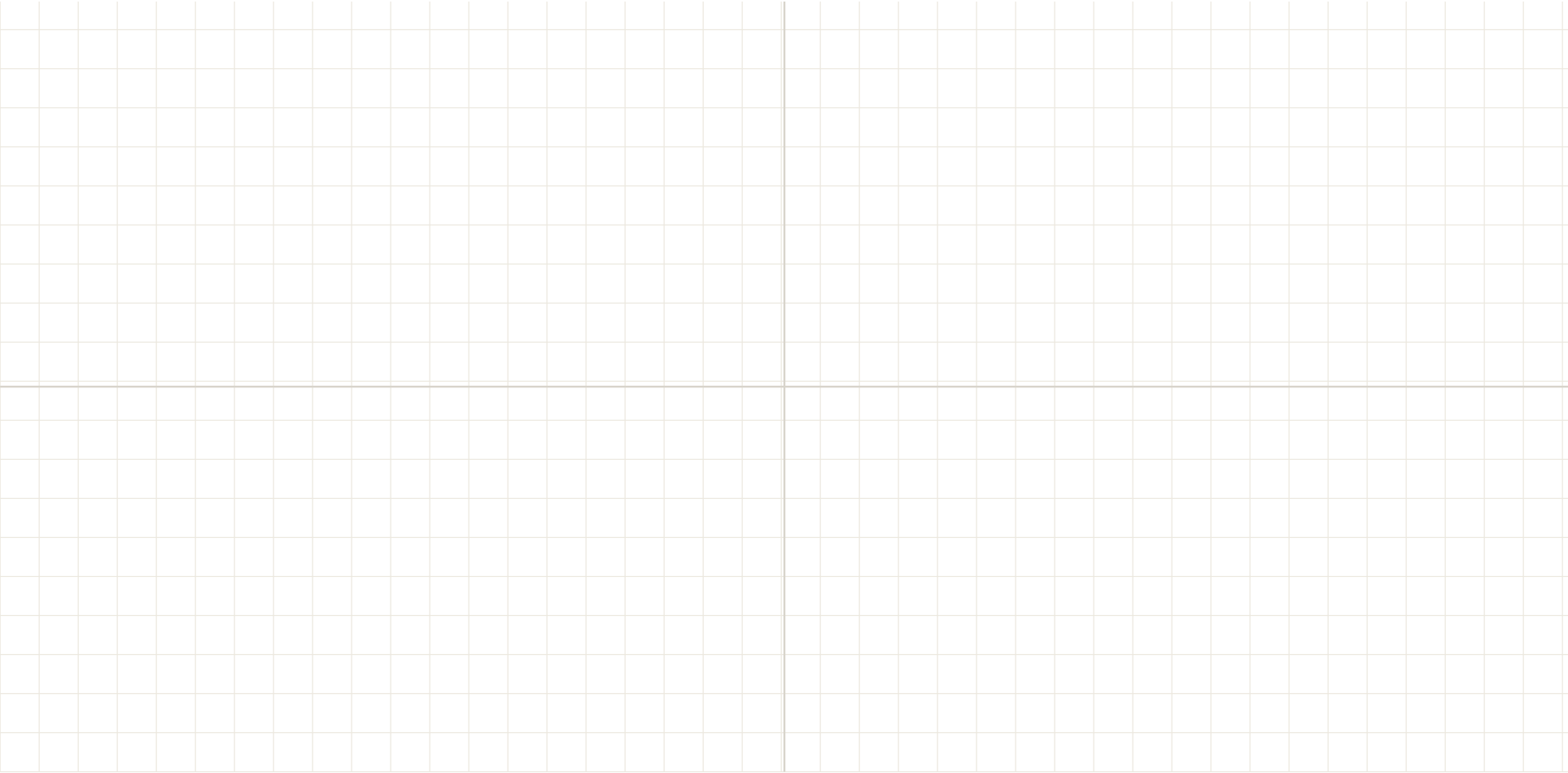
Explicit  
Can you predict term n directly?

explicit  $a_n = a_1 r^{n-1}$

Accumulation  
What does the partial sum represent?

series  $S_n = a_1 + a_2 + \dots + a_n$

OPEN WORKSPACE



# Exponentials + Thresholds

Use this page to test ratios, solve thresholds, interpret parameters, and name where exponential reasoning fails.

| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

Ratio Test  
Are the ratios stable enough?

model  $f(x) = a \cdot b^x$

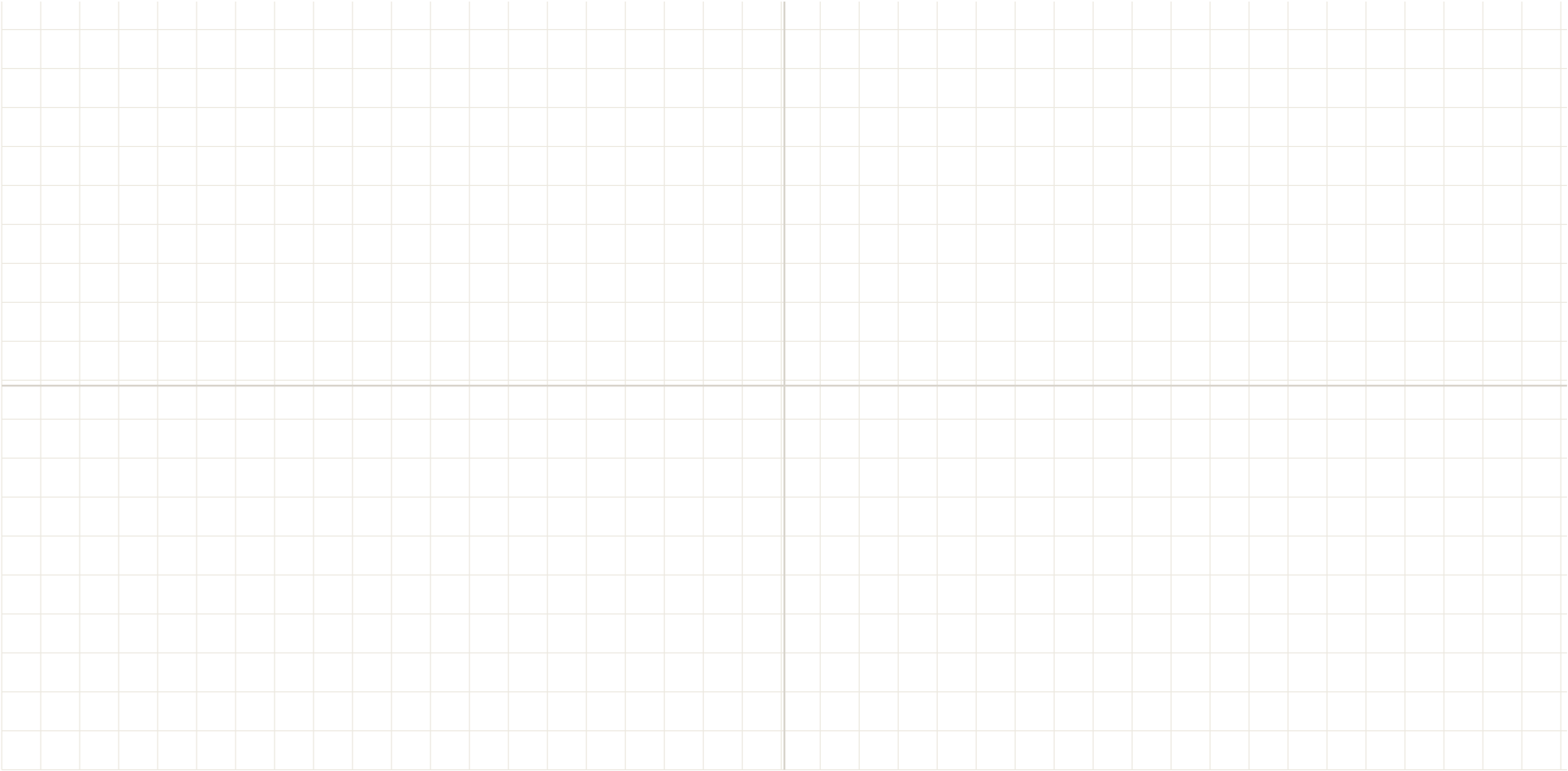
Threshold  
What value matters in the system?

solve  $\log_b(\text{threshold}/a) = x$

Failure  
Where does growth/decay become unrealistic?

meaning  $a = \text{initial value, } b = \text{growth/decay factor}$

OPEN WORKSPACE



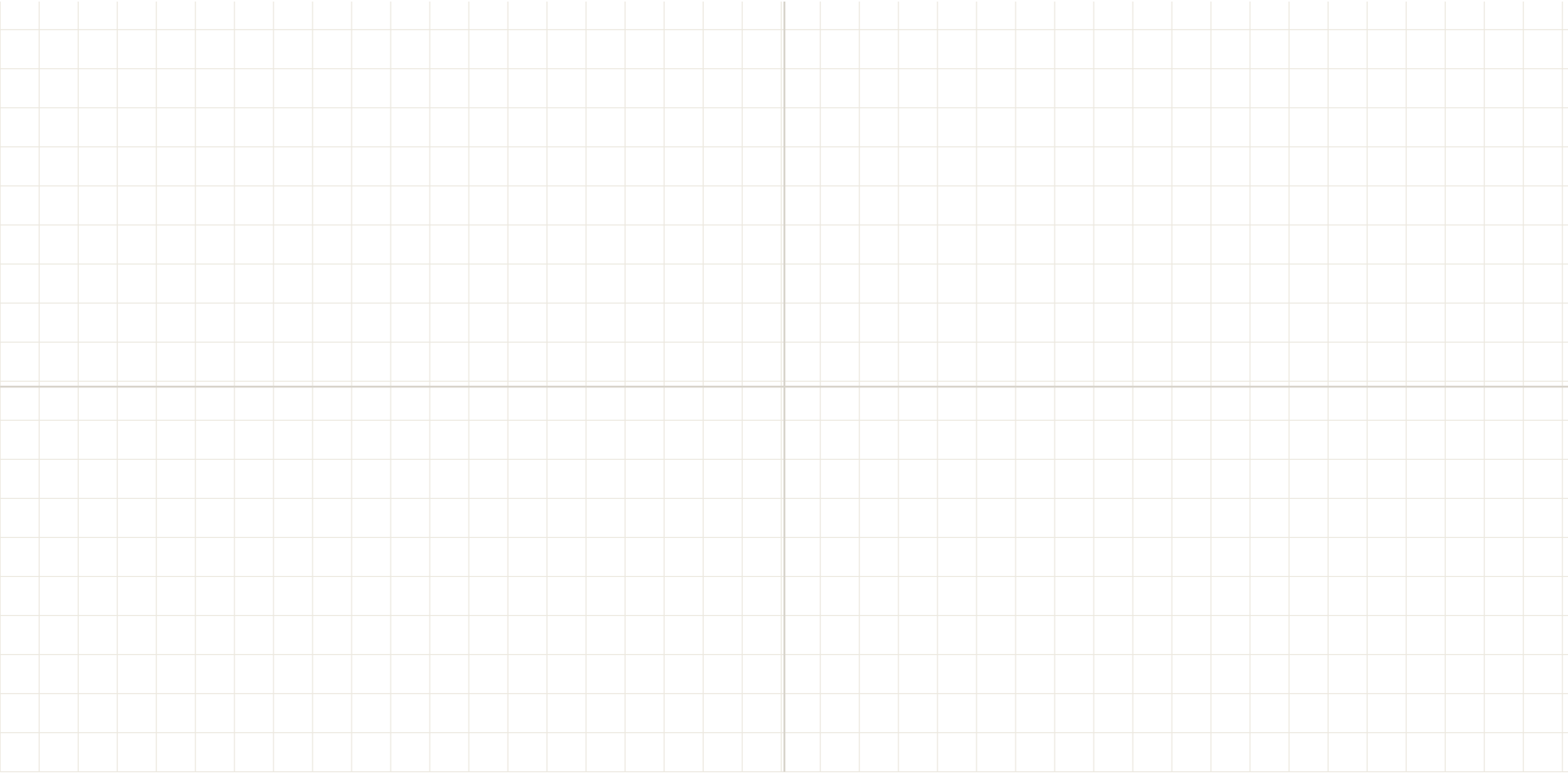
# Polynomial Behavior

Use this page to interpret local shape, turning points, zeros, thresholds, multiplicity, and end behavior realism.

| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

|                                                                                                                                               |                                                                                                                            |                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Local Model</div> <div>What interval justifies this shape?</div> <div>factored</div> <div><math>P(t) = a(t-r_1)(t-r_2)^2...</math></div> | <div>Turning Point</div> <div>What does the reversal mean physically?</div> <div>zero</div> <div><math>P(t)=0</math></div> | <div>End Behavior</div> <div>Where does the polynomial stop making sense?</div> <div>feature</div> <div>multiplicity: cross or bounce?</div> |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|

OPEN WORKSPACE



Use this page to compare models by what they predict, reveal, hide, assume, and fail to explain.

## OPEN WORKSPACE

## Confounding Variables

Use this page to separate visible correlation from possible causes and alternative explanations.

| SYSTEM                                                           | VARIABLES                                                   | MODEL                                                                                                                             | CRITIQUE |
|------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| <div>Related</div> <div>What evidence suggests connection?</div> | <div>Not Proof</div> <div>What else could explain it?</div> | <div>Sentence Frame</div> <div>These variables appear related because ____, but this does not prove causation because ____.</div> |          |
| OPEN WORKSPACE                                                   |                                                             |                                                                                                                                   |          |

Use this page only after your group thinks first: record what Sparky suggested, what you verified, and what you rejected.

## OPEN WORKSPACE

Use this page to show how evidence, critique, or model failure changed your group's interpretation.

## OPEN WORKSPACE

Use this page to make a defensible claim with graph evidence, calculation evidence, rejected-model evidence, and uncertainty.

## OPEN WORKSPACE

Use this page to prepare a mathematical argument: strongest evidence, biggest limitation, rejected model, and final defense.

| Explain                            | Defend                                                              | Close                                   |
|------------------------------------|---------------------------------------------------------------------|-----------------------------------------|
| Every graph and equation verbally. | Mathematical, realism, assumption, interval, counterargument tests. | Not a summary. A mathematical argument. |

Use this page for last annotations, slide notes, and questions your group must be ready to answer.

|                                                   |                                                                         |                                                                       |
|---------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Ready</b><br>All members can explain the math. | <b>Artifacts</b><br>Notebook, vlogs, extensions, and sources are ready. | <b>Attend</b><br>Presentation day is during class on Tuesday, June 9. |
|---------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------|